Customer Segmentation Using K-Means Clustering

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Abstract

This project focuses on segmenting mall customers using clustering techniques. The problem is treated as an unsupervised learning task because the dataset does not include predefined labels, and the goal is to discover natural groupings within the data. By analyzing customer characteristics, the objective is to identify distinct customer segments that can be used to improve marketing strategies.

The dataset used in this analysis is the Mall Customer Segmentation dataset from Kaggle. It includes customer information such as age, gender, annual income, and spending score. These variables are used to understand customer behavior and purchasing patterns. The key variables selected for clustering are age, annual income, and spending score, as they provide meaningful insight into customer differences.

The k-means clustering algorithm was applied to group customers into segments. The optimal number of clusters was determined using the elbow method and silhouette analysis, which suggested that five clusters provide a good representation of the data. The results show that customers can be grouped into distinct segments based on income and spending behavior, such as high-income high-spending customers and low-income low spending customers.

These findings demonstrate how clustering can be used to identify target customer groups and support data driven marketing strategies.

Introduction

This report focuses on analyzing a mall customer dataset to identify distinct customer segments using clustering techniques. The dataset used in this analysis is the Mall Customer Segmentation dataset obtained from Kaggle. It contains information about customers such as demographic characteristics and spending behavior. Unlike classification problems, this dataset does not include a predefined response variable. Instead, the goal is to group customers based on similarities in their attributes, allowing patterns to emerge naturally from the data.

The motivation for this analysis comes from the importance of customer segmentation in business decision-making. Companies often serve a wide range of customers with different behaviors, preferences, and purchasing patterns. Without proper segmentation, it is difficult to design effective marketing strategies. By grouping customers into meaningful segments, businesses can better understand their audience, target specific groups more effectively, and improve overall customer satisfaction.

This problem is appropriate to treat as an unsupervised learning task because there are no labeled outcomes indicating which group each customer belongs to. Unsupervised learning methods, such as clustering, are used to discover hidden structures in the data. Clustering techniques are widely used in marketing analytics to identify customer segments and support strategic decision-making.

The primary analytical question guiding this study is: Can customer demographic and spending behavior data be used to identify meaningful customer segments? To answer this question, the k-means clustering algorithm is applied to the dataset. This method is chosen because it is simple, efficient, and commonly used for customer segmentation tasks.

The experimental setup involves preprocessing the data, selecting relevant variables, and applying clustering techniques to group customers. Methods such as the elbow method and silhouette analysis are used to determine the optimal number of clusters. The results are then visualized and interpreted to understand the characteristics of each customer segment.

The remainder of this report is organized as follows. The Data and Methods section describes the dataset, preprocessing steps, and clustering approach used in the analysis. The Results section presents the clustering outcomes, evaluation methods, and visualizations. Finally, the Conclusion and Discussion section summarizes the key findings, discusses limitations, and suggests directions for future work.

Data and Data preparation

Data Description

The dataset used in this analysis is the Mall Customer Segmentation dataset obtained from Kaggle. This dataset was created for learning purposes and represents customer information collected through a supermarket membership program. It contains basic demographic and behavioral attributes that can be used to understand customer purchasing patterns and support segmentation analysis.

The dataset includes 200 observations and 5 variables. These variables describe different aspects of each customer, including demographic characteristics and spending behavior. Some variables are numeric, representing measurable values, while others are categorical, representing group membership.

The main attributes included in the dataset are described below:

Attribute	Description	Data Type	Range/Categories
CustomerID	Unique identifier for each customer	Numeric	1-200
Gender	Gender of the customer	Categorial	Male, Female
Age	Age of the customer	Numeric	18-70
Annual Income	Customer yearly income(in thousands)	Numeric	15-137k
Spending Score	Score assigned based on spending behavior	Numeric	1-100

Table 1: Description of dataset attributes

For the clustering analysis, the variables Age, Annual Income, and Spending Score were selected. These variables are directly related to customer behavior and are useful for identifying meaningful groups within the data. The CustomerID variable was excluded because it does not provide analytical value, and Gender was not included in clustering to keep the model focused on numerical features.

Data Preparation

Before performing the clustering analysis, several preprocessing steps were applied to prepare the dataset. First, the data was imported into R and inspected using functions such as `head()`, `str()`, and `summary()` to understand its structure and variable types. This step confirmed that the dataset contains 200 observations and does not include missing values.

Next, the Gender variable was converted into a factor to ensure it is properly recognized as a categorical variable. Although this variable was not used in clustering, converting it ensures consistency in the dataset.

For the clustering process, only numeric variables were selected, specifically Age, Annual Income, and Spending Score. These variables were then standardized using scaling techniques. Scaling is necessary because k-means clustering is distance-based, and variables measured on different scales can disproportionately influence the results.

No observations were removed from the dataset, as there were no missing values or extreme outliers that would significantly affect the clustering process. These preprocessing steps ensured that the dataset was clean, consistent, and suitable for applying clustering algorithms.

Methods

This analysis uses a combination of exploratory data analysis (EDA), data preprocessing techniques, and clustering methods to identify patterns in customer behavior. The purpose of these methods is to understand the structure of the dataset, prepare the data for clustering, and evaluate the quality of the resulting clusters.

Exploratory Data Analysis and Descriptive Statistics

Exploratory data analysis was first conducted to examine the structure and distribution of the dataset. Summary statistics were used to understand key variables such as age, annual income, and spending score. Measures such as minimum, maximum, mean, and quartiles were examined to identify general patterns and potential anomalies in the data.

Visualizations, including histograms, scatter plots, and boxplots, were used to better understand the distribution of variables and relationships between them. For example, the scatter plot of annual income versus spending score helped reveal potential groupings of customers, suggesting that clustering techniques would be appropriate for this dataset. Boxplots were also used to check for outliers that could affect clustering results.

Data Preprocessing

Several preprocessing steps were applied to prepare the dataset for clustering. First, relevant variables were selected, including Age, Annual Income, and Spending Score, as these variables capture important aspects of customer behavior.

Next, the selected variables were standardized using scaling techniques. This step is necessary because k-means clustering is based on distance calculations, and variables with larger scales can dominate the clustering process. By scaling the data, all variables contribute equally to the analysis.

No missing values were present in the dataset, so no observations needed to be removed. These preprocessing steps ensured that the data was clean and properly formatted for clustering.

Clustering Method: K-Means

The primary method used in this analysis is the k-means clustering algorithm. K-means is a partitioning method that divides the dataset into k clusters by minimizing the within cluster sum of squares (SSE). Each observation is assigned to the cluster with the nearest centroid, and the centroids are updated iteratively until convergence.

K-means was selected because it is simple, efficient, and widely used for customer segmentation problems. It is particularly effective when the goal is to identify groups of similar observations based on numerical features.

Parameter Selection and Evaluation

To determine the optimal number of clusters, two evaluation methods were used: the elbow method and the silhouette coefficient.

The elbow method evaluates the within cluster sum of squares for different values of k. As the number of clusters increases, the total within cluster variation decreases. The optimal number of clusters is identified at the point where the rate of decrease slows down significantly, forming an "elbow" in the plot.

The silhouette coefficient measures how similar an observation is to its own cluster compared to other clusters. The value ranges from -1 to 1, where higher values indicate better-defined clusters. The average silhouette width was calculated for different values of k to assess cluster quality.

Based on both the elbow method and silhouette analysis, $k = 5$ was selected as the final number of clusters. This choice provides a balance between model simplicity and cluster quality.

Methods Considered but Not Used

One method that was considered but not used in this analysis is hierarchical clustering. Hierarchical clustering can provide a visual representation of cluster structure through dendrograms. However, it was not selected because k-means is more efficient for larger datasets and provides clearer segmentation results for this type of problem.

Another method that was considered but not used is DBSCAN. DBSCAN is useful for identifying clusters of arbitrary shape and detecting noise. However, it requires careful tuning of parameters such as epsilon and minimum points, and it is less suitable for datasets where clusters are relatively well-separated, as in this case.

Key Assumptions

The k-means algorithm relies on several assumptions. It assumes that clusters are spherical in shape and that observations within a cluster are similar to each other. It also assumes that distance is an appropriate measure of similarity and that all variables contribute equally after scaling.

In this dataset, these assumptions are reasonably satisfied. However, real-world customer behavior may not always form perfectly spherical clusters, which is a limitation of the method. Despite this, k-means remains effective for identifying general patterns in the data.

Results

Clustering Results and Evaluation

The k-means clustering algorithm was applied to the standardized dataset using Age, Annual Income, and Spending Score as input variables. To determine the optimal number of clusters, both the elbow method and silhouette analysis were used.

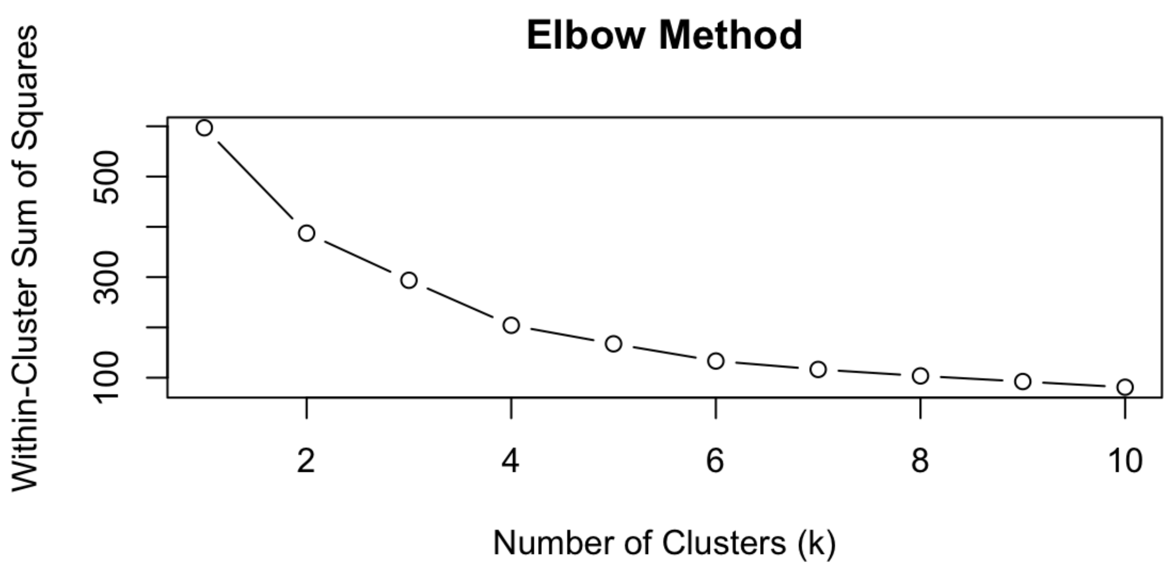

Figure 1: Elbow method for determining the optimal number of clusters.

As shown in Figure 1, the within-cluster sum of squares (SSE) decreases as the number of clusters increases. However, the rate of decrease begins to slow noticeably around k = 5, forming an “elbow” in the plot. This indicates that adding more clusters beyond this point does not significantly improve the model, suggesting that five clusters is a reasonable choice.

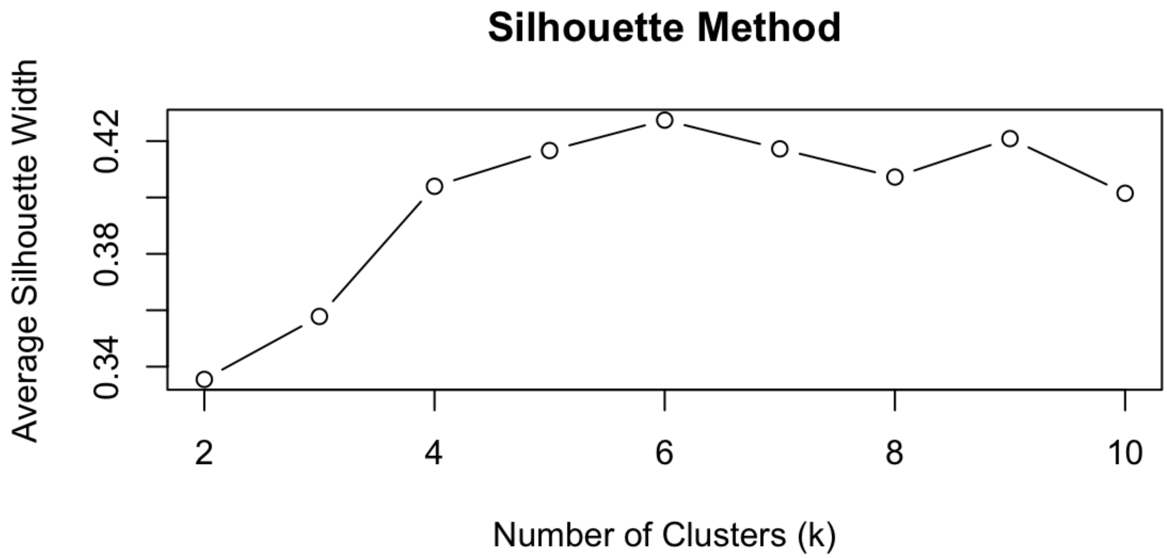

Figure 2: Silhouette analysis for evaluating cluster quality.

Figure 2 shows the average silhouette width for different values of k. The silhouette values are relatively stable and reasonably high around k = 5, indicating that the clusters are well-separated and cohesive. Together, these two methods support the selection of k = 5 for the final clustering model.

Cluster Visualization

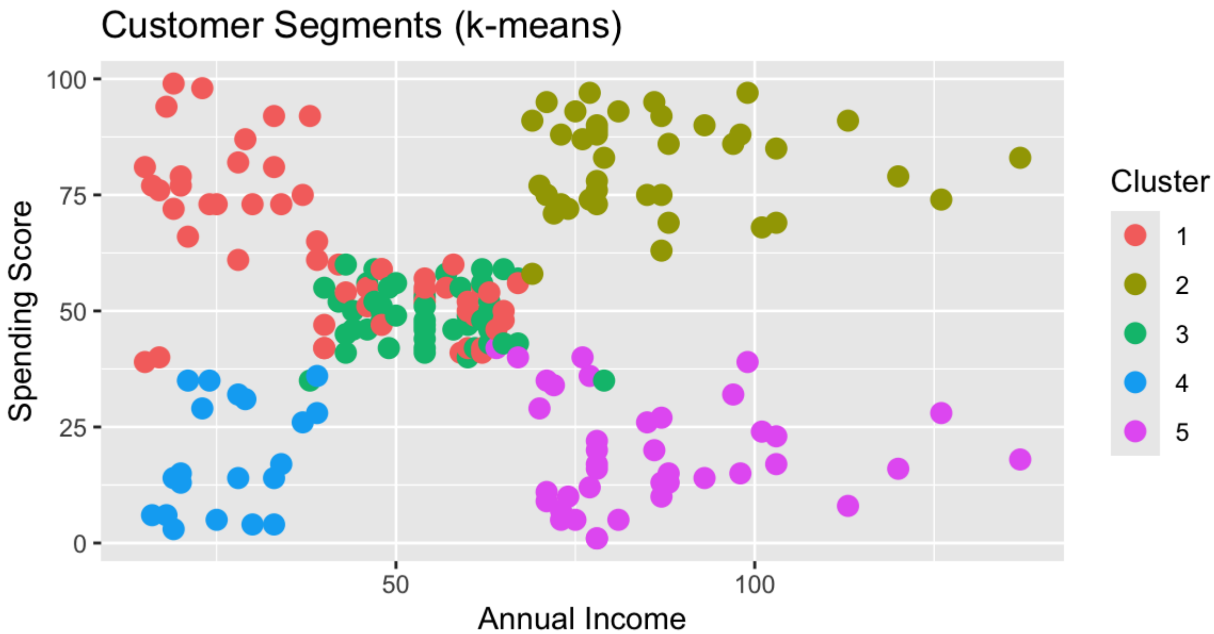

Figure 3: Customer segments based on annual income and spending score.

The clustering results are visualized in Figure 3, which plots annual income against spending score, with each point colored according to its assigned cluster. The plot shows clear separation between clusters, confirming that the algorithm successfully identified distinct groups of customers. The clusters are primarily differentiated based on income level and spending behavior, which are key factors in customer segmentation.

The summary statistics of each cluster provide insight into the characteristics of different customer groups:

- Cluster 2 (High Income, High Spending)
  This cluster has an average income of approximately 86k and a high spending score of about 81.5. These customers are both financially capable and highly engaged in spending. In a real-world business context, this group represents high-value or VIP customers. Companies should prioritize retaining these customers through loyalty programs, exclusive offers, and personalized services, as they contribute significantly to revenue.
- Cluster 5 (High Income, Low Spending)
  Customers in this group also have a high average income (around 86k) but a low spending score (approximately 19.4). This indicates that they have the ability to spend but are not actively purchasing. From a marketing perspective, this group represents a key growth opportunity. Businesses can target these customers with promotions, personalized recommendations, or incentives to encourage increased spending and convert them into high-value customers.

- Cluster 1 (Moderate Income, Active Spending)
  This cluster has an average income of about 41k and a relatively high spending score of 62.2. These customers may not have the highest income, but they are actively engaged in purchasing. In real-world terms, this group represents loyal and consistent customers who contribute steady revenue. Businesses can maintain engagement with this group through discounts, rewards programs, and frequent communication.
- Cluster 4 (Low Income, Low Spending)
  This cluster has the lowest income (around 26.8k) and the lowest spending score (approximately 18.4). These customers are less likely to spend due to limited financial capacity. From a business perspective, this group represents a **low-value segment**, and companies may choose to allocate fewer resources toward targeting them, focusing instead on higher-value segments.
- Cluster 3 (Older, Moderate Behavior)
  Customers in this cluster have the highest average age (around 55.6), with moderate income and spending levels. This suggests a more conservative and stable customer group. In real-world scenarios, these customers may prioritize essential or practical purchases over luxury spending. Businesses may target them with value based or necessity driven products.

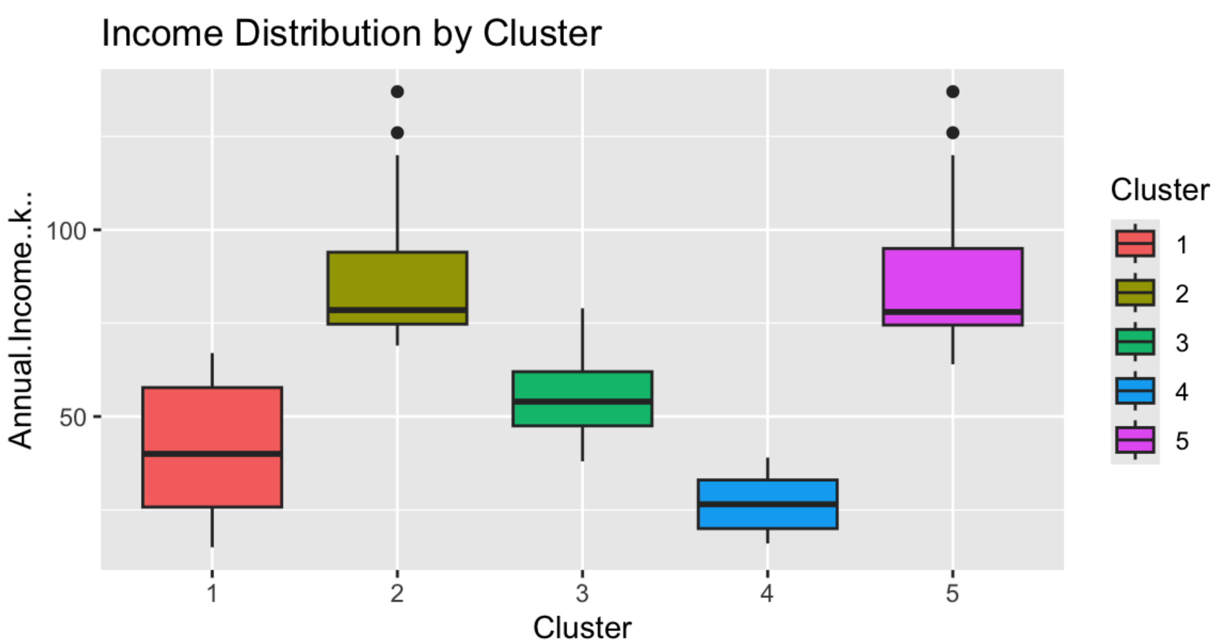

Figure 4: Distribution of annual income across clusters.

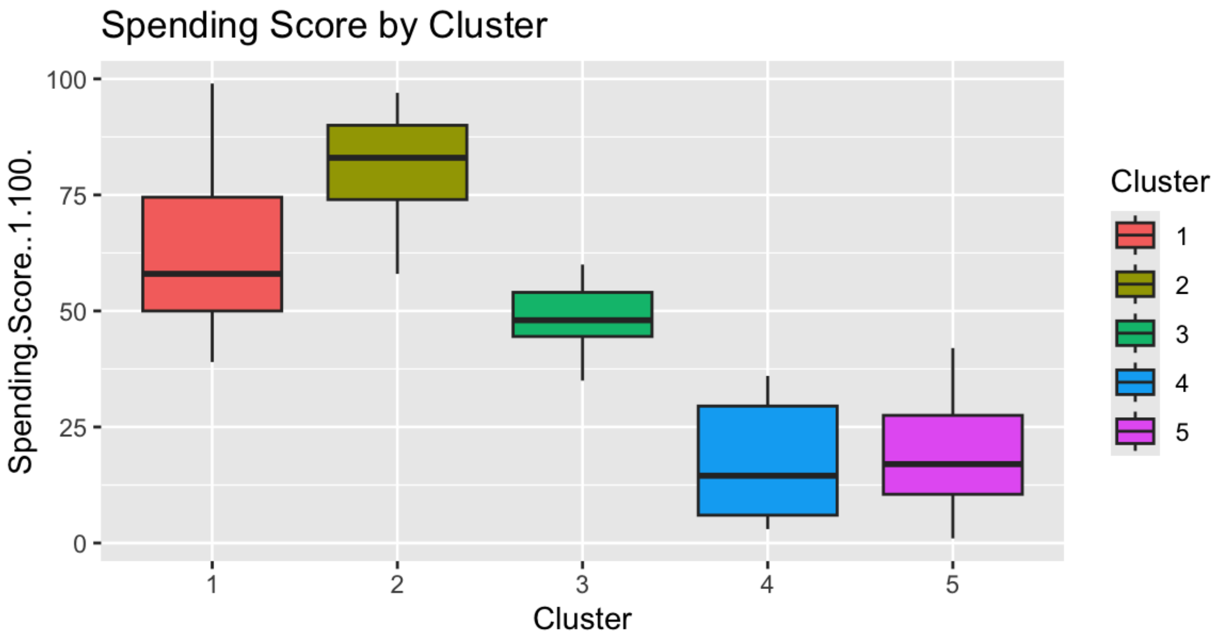

Figure 5: Distribution of spending score across clusters.

The boxplots in Figures 4 and 5 further highlight the differences between clusters. Figure 4 shows that income levels vary significantly across clusters, clearly separating high-income and low-income groups. Figure 5 shows similar variation in spending behavior, with some clusters having consistently high spending scores while others have low scores.

These visualizations confirm that the clusters represent meaningful differences in customer characteristics rather than random groupings.

The clustering analysis successfully identified five distinct customer segments based on income, spending behavior, and age. These segments provide valuable insights into customer behavior and can be used to guide business decisions.

From a real-world perspective, these results allow businesses to move beyond a one-size-fits-all approach and instead apply targeted marketing strategies. For example, focusing on high-income low-spending customers can increase revenue, while maintaining engagement with high-spending customers can improve long-term retention.

Conclusion

This analysis focused on segmenting mall customers using clustering techniques. The dataset was explored and preprocessed to ensure it was suitable for analysis, and the k-means clustering algorithm was applied to group customers based on age, annual income, and spending score. The optimal number of clusters was determined using the elbow method and silhouette analysis, which both suggested that five clusters provide a good representation of the data.

The results show that the dataset can be divided into five distinct customer segments, each with different characteristics in terms of income, spending behavior, and age. These clusters provide meaningful insights into customer behavior. For example, high-income high-spending customers represent the most valuable segment, while high-income low-spending customers represent a key opportunity for increasing revenue through targeted marketing strategies. Other groups, such as low-income low-spending customers and older customers with moderate behavior, reflect different levels of engagement and purchasing patterns.

One of the main strengths of this analysis is that it demonstrates how unsupervised learning can uncover hidden patterns in data without the need for predefined labels. By grouping customers based on similarity, businesses can better understand their audience and design more effective marketing strategies. Instead of treating all customers the same, companies can tailor their approach to each segment, which can improve customer satisfaction and overall business performance.

However, there are several limitations to this analysis. First, only three variables were used for clustering, which may not fully capture all aspects of customer behavior. Additional variables, such as purchase history or frequency of visits, could provide deeper insights. Second, k-means assumes that clusters are spherical and equally sized, which may not always reflect real-world customer patterns. Third, the results are sensitive to scaling and the choice of the number of clusters, which can affect the final segmentation.

These limitations impact how the results should be interpreted. While the clusters provide useful insights, they may not fully represent all customer behaviors, and different methods or variables could produce different groupings. Therefore, the results should be viewed as a general representation of customer segments rather than exact classifications.

Future work could improve this analysis in several ways. One possible improvement is to include additional variables that better capture customer behavior, such as transaction frequency or product preferences. Another approach would be to apply different clustering methods, such as hierarchical clustering or DBSCAN, to compare results and identify alternative patterns. Dimensionality reduction techniques, such as principal component analysis (PCA), could also be used to improve visualization and clustering performance.

Overall, this analysis demonstrates how clustering techniques can be used to segment customers and provide actionable insights for business decision-making. The results highlight the importance of understanding customer behavior and show how data-driven approaches can support more effective marketing strategies.

Reference

Choudhary, V. (2019). *Mall customer segmentation dataset*. Kaggle. https://www.kaggle.com/datasets/vjchoudhary7/customer-segmentation-tutorial-in-python

Dolnicar, S., Grün, B., & Leisch, F. (2018). *Market segmentation analysis: Understanding it, doing it, and making it useful*. Springer. https://doi.org/10.1007/978-981-10-8818-6

Jain, A. K. (2010). Data clustering: 50 years beyond k-means. *Pattern Recognition Letters, 31*(8), 651–666. https://doi.org/10.1016/j.patrec.2009.09.011

Likas, A., Vlassis, N., & Verbeek, J. J. (2003). The global k-means clustering algorithm. *Pattern Recognition, 36*(2), 451–461. https://doi.org/10.1016/S0031-3203(02)00060-2

Xu, R., & Wunsch, D. (2005). Survey of clustering algorithms. *IEEE Transactions on Neural Networks, 16*(3), 645–678. https://doi.org/10.1109/TNN.2005.845141